

ESTIMATING TREATMENT EFFECTS WHEN SUBJECTS LEARN FROM THE PRETEST

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Summary: The Solomon four-group design is employed in psychological research when the effect of a training, education or treatment program is assessed through a test or questionnaire. The Solomon four-group design combines paired (pre-test and post-test) and unpaired (post-test only) collection of data and therefore allows one to assess both a possible pre-test sensitization effect, and the treatment effect. The statistical analysis of binary data from a Solomon four-group design can be based on the joint likelihood of pre- and post-test scores. Effects of interest, like the pre-test sensitization and treatment effects are assessed through a standard Analysis of Deviance of a generalized linear model for the conditional probabilities of the post-test scores, given the pre-test scores. Thus the analysis of binary data from a Solomon four-group design is conceptually simpler than the analysis proposed by Braver and Braver (1988) for continuous data. This research was motivated by data from an evaluation study of an AIDS education and training program for employees of a supermarket chain in Cape Town, South Africa.

INTRODUCTION

In psychological research it is a frequent goal to assess the effect of a training, education or treatment program through some test or questionnaire. The most simple design for that purpose is the post-test only control group design (table 1). Here one randomises a number of experimental subjects to

one of two groups, such that one group receives the treatment program while the other group receives no treatment. After completion of the treatment program a suitable test (post-test) is applied to measure the responses of the subjects in both groups. A comparison of the test scores of the two groups would then allow an assessment of the treatment effect.

Because of the often great inter-subject variation in the test results it would be desirable to also apply a test before the treatment program (pre-test), which would allow one to "adjust for baseline", and thus to reduce the effect of inter-subject variation on the assessment of the treatment effect (see the pre- and post-test control group design in table 1). However, the very fact that a pre-treatment test is applied to the experimental subjects might constitute a treatment or training, or might sensitize the subjects to the treatment program under consideration, so that this so-called pre-test sensitization effect might bias the estimation of the treatment effect. Further, in such research the pre-test sensitization effect may in itself be an effect of interest.

Solomon (1949) proposed an experimental design which allows one to assess both the effect of a treatment and the pre-test sensitization effect. The Solomon four-group design combines the pre- and post-test control group design, and the post-test only control group design, so that experimental subjects are randomly allocated to one of four groups: in group 1, a pre-test is applied, followed by a treatment and a post-test. In group 2, only the pre- and post-test are applied, but no treatment. In group 3, no pre-test, but treatment and a post-test are applied, and in group 4, only the post-test is applied. The design is shown in table 1. Braver and Braver (1988) give a thorough review of the history and the rationale of the Solomon four-group design. The general goal of the analysis is the assessment of the treatment effect, but Braver and Braver (1988, p. 150) point out that the Solomon four-group design is the only one of three one-treatment designs (the other two being the pre- and post-test control group design, and the post-test only control group design shown on table 1) which allows one to assess the presence of the pre-test sensitization effect. They propose a

method for the statistical analysis of data from a Solomon four-group design, concentrating on the case of continuous data. Their analysis involves Analysis of Variance and Analysis of Covariance techniques with at least a tacit appeal to normal theory results.

Table 1. Three one-treatment experimental designs

Design	Group	Pre-test	Treatment	Post-test
Solomon four-group	1	Yes	Yes	Yes
	2	Yes	No	Yes
	3	No	Yes	Yes
	4	No	No	Yes
Pre- and post-test control group	1	Yes	Yes	Yes
	2	Yes	No	Yes
Post-test only control group	1	No	Yes	Yes
	2	No	No	Yes

In the present paper the statistical analysis of *binary* data from a Solomon four-group design is discussed. Here the pre- and post-tests do not yield continuous measurements, but scores of either 1 or 0 (say). Braver and Braver (1988) point out that in the case of continuous data it is not trivial to combine efficiently the pre-test information from groups 1 and 2, the post-test information from groups 1 and 2, and the post-test information from groups 3 and 4, for inference on the treatment effect. In the case of binary data, a joint likelihood for all the data, that is, the pre- and post-test scores for all groups, can be specified. This involves the specification of a generalized linear model (McCullagh & Nelder, 1983) for the conditional probabilities, given the pre-test score, of having post-test score 1. The analysis of effects of interest, like the treatment and pre-test sensitization effects, can be performed through a standard Analysis of Deviance (McCullagh & Nelder, 1983, pp. 26-28) of this generalized linear model.

Thus the analysis of binary data from a Solomon four-group design is conceptually simpler than that proposed by Braver and Braver (1988) for continuous data, even though some computational effort is required to maximize the likelihood. In particular, because the analysis is based on the joint likelihood for all the data, it is not necessary to perform two separate analyses of the data from groups 1 and 2, and groups 3 and 4 respectively, and to combine the results through a meta-analytic approach as suggested by Braver and Braver (1988) for continuous data.

The present research was motivated by data from an evaluation study of an AIDS education and training program for employees of a supermarket chain in Cape Town, South Africa. Subjects underwent a pre-training interview, a training and education program on AIDS, and a post-training interview according to the specifications of a Solomon four-group design. The pre- and post-training interviews consisted of the same set of knowledge questions about AIDS, and for each question the subjects were scored according to whether or not they gave the correct answer to the question. Data from this study will be used to illustrate the methodology described below.

2. STATISTICAL ANALYSIS

2.1 The data

Binary data from a Solomon four-group design is of the following form: In groups 1 and 2, x_{1j} out of n_{1j} subjects have pre-test score 1 (see the row marginals in table 2a). In all four groups, y_{ij} out of n_{ij} subjects have post-test score 1 (see the column marginals in table 2a, and table 2b).

On the basis of the pre-test score, the subjects in groups 1 and 2 can further be stratified as in table 2a: z_{1j1} of the x_{1j} subjects who had pre-test score 1 have post-test score 1, and z_{1j2} out of the $n_{1j} - x_{1j}$ have post-test score 1.

Table 2a. Data from groups 1 and 2

		Post-test		
		1	0	Total
Pre-test	1	z_{1j1}	$x_{1j} - z_{1j1}$	x_{1j}
	0	z_{1j2}	$n_{1j} - x_{1j} - z_{1j2}$	$n_{1j} - x_{1j}$
Total		y_{1j}	$n_{1j} - y_{1j}$	n_{1j}

Table 2b. Data from groups 3 and 4

		Post-test		
		1	0	Total
		y_{2j}	$n_{2j} - y_{2j}$	n_{2j}

2.2 The model

Let p be the probability that a subject has pre-test score 1, so that $(1 - p)$ is the probability that a subject has pre-test score 0. Let the conditional probability, given the pre-test score, of having post-test score 1 be q_{ijk} , where $i = 1, 2$ indicates whether the subject underwent a pre-test, $j = 1, 2$ indicates whether the subject underwent the treatment, and $k = 1, 2$ indicates whether the pre-test score is 1 or 0. Note that the conditional probabilities q_{2jk} for the subjects who did not undergo a pre-test cannot be estimated directly from the data, but for the moment this is unimportant. Thus the q_{ijk} can be organized in a three-way layout with factors 'pre-test' (present/absent), 'treatment' (present/absent) and 'pre-test score' (1 or 0). The following generalized linear model (McCullagh & Nelder, 1983) for the conditional probabilities q_{ijk} is postulated

$$g(q_{ijk}) = \mu + \alpha_i + \tau_j + \gamma_k + (\alpha\tau)_{ij} + (\alpha\gamma)_{ik} + (\tau\gamma)_{jk} + (\alpha\tau\gamma)_{ijk} \quad (1)$$

where $g(\cdot)$ is a suitable link function (McCullagh & Nelder, 1983, pp. 23-34), for example the logit. Here μ denotes the grand mean, α , τ and γ denote the main effects due to 'pre-test' (P), 'treatment' (T) and 'pre-test score' (PS), $(\alpha\tau)_{ij}$, $(\alpha\gamma)_{ik}$, and $(\tau\gamma)_{jk}$ the P×T, P×PS and T×PS two-way interactions, and $(\alpha\tau\gamma)_{ijk}$ the P×T×PS three-way interaction in the usual notation for a three-way layout.

Because the pre-test information is not available for groups 3 and 4 of the Solomon four-group design, two effects in the model of equation 1 are not estimable. These are the P×PS two-way interaction, and the P×T×PS three-way interaction. Thus the largest model actually fitted to the data is

$$g(q_{ijk}) = \mu + \alpha_i + \tau_j + \gamma_k + (\alpha\tau)_{ij} + (\tau\gamma)_{jk}. \quad (2)$$

The model of equation 2 is overparameterized in the sense that fifteen parameters on the right-hand side of equation (2) are fitted to model the eight probabilities q_{ijk} . Applying the usual summation constraints on the terms in equation 2 (see appendix), equation 2 can be reparameterized and written concisely as

$$g(q) = X\beta \quad (3)$$

where $q = (q_{ijk})$ is an 8×1 column vector, $X_{8 \times 6}$ is a full rank design matrix, and $\beta_{6 \times 1}$ is the column parameter vector associated with the grand mean, the three main effects, and the two estimable two-way interactions. The design matrix X is given in appendix 1.

The model for the data is fully specified by the parameter p , that is, the marginal probability of having pre-test score 1, and equations 2 or 3, which specify the conditional probabilities of having post-test score 1 in the different strata.

2.3 The likelihood

The model of equation 2, and any submodel thereof, can be fitted by maximum likelihood. The likelihood can be split into three components. The first component is the likelihood associated with the marginal

distribution of the pre-test scores in groups 1 and 2, because the x_{1j} follow binomial distributions $B(n_{1j}, p)$, where n_{1j} denotes the sample size and p the probability. The second component is the likelihood associated with the conditional distribution of the post-test scores in group 1 and 2, because the z_{1j1} and z_{1j2} follow, conditionally on the pre-test scores, binomial distributions $B(x_{1j}, q_{1j1})$ and $B(n_{1j} - x_{1j}, q_{1j2})$. Finally, noting that the marginal probabilities r_{ij} of having post-test score 1 in the four groups of the Solomon four-group design are

$$r_{ij} = pq_{ij1} + (1 - p)q_{ij2}, \quad i, j = 1, 2 \quad (4)$$

the third component of the likelihood is the likelihood associated with the marginal distribution of the post-test scores in groups 3 and 4, because the y_{2j} follow binomial distributions $B(n_{2j}, r_{2j})$. Thus the three components of the log-likelihood are

$$\begin{aligned} l_1 &= \sum_j \left[x_{1j} \log \frac{p}{1-p} + n_{1j} \log(1-p) + \log \left[\binom{n_{1j}}{x_{1j}} \right] \right], \\ l_2 &= \sum_j \left[z_{1j1} \log \frac{q_{1j1}}{1-q_{1j1}} + x_{1j} \log(1-q_{1j1}) + \log \left[\binom{x_{1j}}{z_{1j1}} \right] \right], \\ &\quad + \sum_j \left[z_{1j2} \log \frac{q_{1j2}}{1-q_{1j2}} + (n_{1j} - x_{1j}) \log(1-q_{1j2}) + \log \left[\binom{n_{1j} - x_{1j}}{z_{1j2}} \right] \right], \text{ and} \\ l_3 &= \sum_j \left[y_{2j} \log \frac{r_{2j}}{1-r_{2j}} + n_{2j} \log(1-r_{2j}) + \log \left[\binom{n_{2j}}{y_{2j}} \right] \right]. \end{aligned}$$

Note that the r_{ij} are a function of p and q_{ijk} through equation 4, and the q_{ijk} are in turn a function of β through equation 3. The joint log-likelihood for the data is

$$l = l(p, \beta) = l_1 + l_2 + l_3.$$

This likelihood can be maximized using Fisher's method of scoring (Dobson, 1983, pp. 30-32) as described in appendix 2. Under the "full" model of equation 2, seven independent parameters, namely p , and the six

components of β which correspond to the grand mean, the P, T and PS main effects, and the P×P and T×PS interactions are estimated.

2.4 The analysis

The analysis of the data is through an Analysis of Deviance (McCullagh & Nelder, 1983, pp. 26–28) of the three-way layout of equation 2. Two effects are of main interest: the P×T interaction, which represents the effect of pre-test sensitization, and the ‘treatment’ main effect. In fact, a third interesting effect can be tested, namely the T×PS interaction. The T×PS term allows for the fact that the treatment effect may depend on the pre-test score. (The problem of testing this term was not addressed by Braver and Braver (1988).) A possible sequence of nested models to fit by maximum likelihood would therefore be

$$g(q_{ijk}) = \mu + \alpha_i + \tau_j + \gamma_k + (\alpha\tau)_{ij} + (\tau\gamma)_{jk}, \quad (5)$$

$$g(q_{ijk}) = \mu + \alpha_i + \tau_j + \gamma_k + (\alpha\tau)_{ij}, \quad (6)$$

$$g(q_{ijk}) = \mu + \alpha_i + \tau_j + \gamma_k, \text{ and} \quad (7)$$

$$g(q_{ijk}) = \mu + \alpha_i + \gamma_k. \quad (8)$$

Equation 5 specifies the “full” model of equation 2, in equation (6) the T×PS interaction is dropped, in equation 7 the P×T interaction is also dropped, leaving only the main effects in the model, and in equation 8 the ‘treatment’ main effect is dropped. If $l_{\max}(5)$ denotes the maximized likelihood for the model of equation 5, and similarly for the other models, then $D_{(5)} = 2[l_{\max}(5) - l_{\max}(5)] = 0$ is the deviance (McCullagh & Nelder, 1983, pp. 24–26) for the model of equation 5, $D_{(6)} = 2[l_{\max}(5) - l_{\max}(6)]$ is the deviance for the model of equation 6, $D_{(7)} = 2[l_{\max}(5) - l_{\max}(7)]$ is the deviance for the model of equation 7, and finally $D_{(8)} = 2[l_{\max}(5) - l_{\max}(8)]$ is the deviance for the model of equation 8. Thus one has a sequence of deviances associated with a sequence of nested models. The first difference of this sequence of deviances yields likelihood ratio test statistics for effects of

interest. Specifically, $D_{(6)} - D_{(5)} = 2[l_{\max}(5) - l_{\max}(6)]$ is a test statistic for the hypothesis of no T*PS interaction, $D_{(7)} - D_{(6)} = 2[l_{\max}(6) - l_{\max}(7)]$ is a test statistic for the hypothesis of no P*T interaction, and $D_{(8)} - D_{(7)} = 2[l_{\max}(7) - l_{\max}(8)]$ is a test statistic for the hypothesis of no 'treatment' effect. Analysis of Deviance tables corresponding to the above sequence of models are given with the example below.

The likelihood ratio test statistics appearing in the Analysis of Deviance tables follow asymptotically a $\chi^2(1)$ distribution under the null-hypothesis that the tested effects are zero. A sample size of 40 is usually considered adequate for the asymptotic distribution of the chi-square test in a single 2*2 table to be valid. Because the data from a Solomon four-group design is effectively summarized in three 2*2 tables (see table 3 below), it is recommended that the total sample size for all 4 groups of the Solomon four-group design is at least 120.

2.5 Quantification of the treatment effect

To quantify the treatment effect, one can use the odds ratio

$$OR_{ik} = \frac{q_{i1k} / (1 - q_{i1k})}{q_{i2k} / (1 - q_{i2k})}, \quad i, k = 1, 2 \quad (9)$$

which is the odds of having post-test score 1 for treated subjects over the odds of having post-test score 1 for untreated subjects. Here OR_{11} is the odds ratio for pre-tested subjects with pre-test score 1, OR_{12} the odds ratio for pre-tested subjects with pre-test score 0, OR_{21} the odds ratio for un-pretested subjects with pre-test score 1, and OR_{22} the odds ratio for un-pretested subjects with pre-test score 0. In terms of the parameter vector $\beta = [\beta_1, \dots, \beta_6]'$ of the full rank reparameterization of the model of equation 5, the log odds ratios can be written as

$$\begin{aligned} \log OR_{11} &= 2(\beta_3 + \beta_5 + \beta_6), \\ \log OR_{12} &= 2(\beta_3 + \beta_5 - \beta_6), \\ \log OR_{21} &= 2(\beta_3 - \beta_5 + \beta_6), \text{ and} \\ \log OR_{22} &= 2(\beta_3 - \beta_5 - \beta_6). \end{aligned}$$

Note that because the model of equation 2 does not fit the $P \times T \times PS$ interaction, it follows that $\log OR_{11} - \log OR_{12} = \log OR_{21} - \log OR_{22}$. If there is no $T \times PS$ interaction and the model of equation 6 is fitted, then $OR_{11} = OR_{12} = OR_1$ and $OR_{21} = OR_{22} = OR_2$, and

$$\log OR_1 = 2(\beta_3 + \beta_5), \text{ and}$$

$$\log OR_2 = 2(\beta_3 - \beta_5).$$

If further there is no $P \times T$ interaction and the model of equation 7 is fitted, then $OR_{11} = OR_{12} = OR_{21} = OR_{22} = OR$, and

$$\log OR = 2\beta_3.$$

The maximum likelihood estimates of the log odds ratios can thus easily be obtained as linear contrasts of the maximum likelihood estimate $\hat{\beta}$ of β . An estimate of the standard error of the maximum likelihood estimates of the log odds ratios can be computed using the expected information matrix H , since any linear contrast $c' \hat{\beta}$ of $\hat{\beta}$ has estimated standard error $SE(c' \hat{\beta}) = \sqrt{c' H^{-1} c}$.

3. EXAMPLE

In an evaluation study of an AIDS education and training program subjects were randomly allocated to the four groups of a Solomon four-group design. According to the specifications of the design, subjects underwent a pre-training interview, an AIDS training and education program, and a post-training interview. The pre- and post-training interviews consisted, among others, of the same set of sixteen knowledge questions about AIDS, which could be answered either 'yes', 'no' or 'don't know'. On each question the subjects were scored according to whether or not they gave the correct answer to the question (if they answered 'don't know', they were scored as having given the wrong answer).

The primary aim of the study was to evaluate the effectiveness of the training program in increasing individuals' knowledge about AIDS. However, prior experience in AIDS education indicated that the interviews conducted to assess pre-training knowledge had educational value in themselves, as individuals were made aware of their lack of knowledge. Thus the Solomon four-group design was chosen for the study because it allows one to evaluate both the effect of the pre-training interview and the effect of the training program.

To illustrate the statistical methodology described above, data on two questions are analysed. The first question was "If a woman with AIDS gets pregnant, is her baby likely to develop AIDS?" Subjects were scored 1 if they answered 'yes', and 0 otherwise. The second question was "Can AIDS be passed on by shaking hands with someone who has AIDS?" Here subjects were scored 1 if they answered 'no', and 0 otherwise. The data for these two questions are given in tables 3 and 4.

Tables 5 and 6 show the Analyses of Deviance of the two data sets. In each table, the first column specifies a sequence of nested models, and the second column the deviances associated with the models. The third column gives the first differences of the deviances, that is, the deviance of a given model is subtracted from the deviance of the next smaller model which has one effect less. The difference in deviances represents a likelihood ratio test statistic for the effect in question. The fourth column gives the (approximate) P-value associated with the test statistic, computed by comparing the observed test statistic with a $\chi^2(1)$ distribution.

For question 1, neither the 'treatment' by 'pre-test score' interaction (approximate $\chi^2(1) = 0,02$, $P = 0,90$) nor the 'pre-test' by 'treatment' interaction (approximate $\chi^2(1) = 1,39$, $P = 0,24$) appears significant. However, the approximate chi-square test statistic for the treatment main effect is $\chi^2(1) = 5,10$ ($P = 0,02$), so that the treatment effect does appear significant. Because the $T \times PS$ and $P \times T$ interactions can be dropped from the model, the treatment effect can be considered to be the same whether or not a subject was pre-tested, and whatever the score of the subject in the

pretest. The odds ratio for treatment vs. no treatment is 3,0 (log odds ratio 1,1; standard error of the log odds ratio 0,52).

Table 3. Data for question 1

		<u>Group 1</u>		<u>Group 2</u>	
		Post-test		Post-test	
		1	0	1	0
Pre-test	1	35	3	43	6
	0	1	1	2	1

		<u>Group 3</u>		<u>Group 4</u>	
		Post-test		Post-test	
		1	0	1	0
		63	2	51	9

Table 4. Data for question 2

		<u>Group 1</u>		<u>Group 2</u>	
		Post-test		Post-test	
		1	0	1	0
Pre-test	1	23	1	22	2
	0	13	6	6	22

		<u>Group 3</u>		<u>Group 4</u>	
		Post-test		Post-test	
		1	0	1	0
		63	2	33	27

Table 5. Analysis of deviance (question 1)

Model	Deviance	First difference of deviances	P-value
P PS	6,51		
P T PS	1,41	5,10	0,02
P T PS P×T	0,02	1,39	0,24
P T PS P×T T×PS	0	0,02	0,90

Table 6. Analysis of deviance (question 2)

Model	Deviance	First difference of deviance	P-value
P PS	45,97		
P T PS	5,71	40,26	<0,00005
P T PS P×T	0,79	4,92	0,03
P T PS P×T T×PS	0	0,79	0,38

For question 2, the 'treatment' by 'pre-test score' interaction is not significant (approximate $\chi^2(1) = 0,79$, $P = 0,38$), but the 'pre-test' by 'treatment' interaction is (approximate $\chi^2(1) = 4,92$, $P = 0,03$). Thus the effect of treatment is different for the pre-tested and the un-pretested groups. In the pre-tested groups, the odds ratio for treatment vs. no treatment is 6,2 (log odds ratio 1,8; standard error 0,61), while in the un-pretested groups the odds ratio is 58,4 (log odds ratio 4,1; standard error 0,92), emphasizing the 'pre-test' by 'treatment' interaction. Approximate z-statistics based on the log odds ratio and its standard error are $z = 2,95$ ($P = 0,0016$) for the pre-tested groups, and $z = 4,46$ ($P < 0,00005$) for the un-pretested groups. Thus there is a significant treatment effect both for the

pre-tested and the un-pretested groups. However, the treatment effect is stronger in the un-pretested groups, a somewhat surprising result as it was expected that pre-test sensitization would, if anything, increase the efficacy of the training program.

4. DISCUSSION

The above example illustrates that the analysis of binary data from a Solomon four-group design is relatively straightforward compared to the analysis of continuous data proposed by Braver and Braver (1988). This is so because, in the case of binary data, the analysis can be based on the joint likelihood for all the data, so that the information from the pre-tested groups can be combined simply and efficiently. Test statistics for effects of interest, like the pre-test sensitization effect and the treatment effect are obtained in a standard manner through likelihood ratio tests, or, in the terminology of generalized linear models, through an Analysis of Deviance. It is also useful that the treatment effect can be quantified through the odds ratio, a standard effect measure for binary data. The odds ratio augments the information provided by the chi-square statistic for treatment, as the chi-square statistic indicates the statistical significance, but the odds ratio indicates the magnitude of the treatment effect.

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APPENDIX 1

The model of equation 3 is a reparameterization of the model of equation 2 obtained through imposing the summation constraints $\sum_i \alpha_i = 0$, $\sum_j \tau_j = 0$, $\sum_k \tau_k = 0$, $\sum_j (\alpha\tau)_{ij} = 0$ for all i , $\sum_i (\alpha\tau)_{ij} = 0$ for all j , $\sum_j (\tau\gamma)_{jk} = 0$ for all k , and $\sum_k (\tau\gamma)_{jk} = 0$ for all j . The design matrix X and the parameter vector β of equation 3 are then given by

$$X = \begin{bmatrix} +1 & +1 & +1 & +1 & +1 & +1 \\ +1 & +1 & +1 & -1 & +1 & -1 \\ +1 & +1 & -1 & +1 & -1 & -1 \\ +1 & +1 & -1 & -1 & -1 & +1 \\ +1 & -1 & +1 & +1 & -1 & +1 \\ +1 & -1 & +1 & -1 & -1 & -1 \\ +1 & -1 & -1 & +1 & +1 & -1 \\ +1 & -1 & -1 & -1 & +1 & +1 \end{bmatrix}$$

and $\beta = [\beta_1, \dots, \beta_6]'$, where β_1 represents the grand mean, β_2 the pre-test (P) effect, β_3 the treatment (T) effect, β_4 the pre-test score (PS) effect, β_5 the P×T interaction, and β_6 the T×PS interaction. Submodels of the "full" model of equation (2), like those submodels specified in equations 6 to

8, are obtained by dropping certain effects from the full model. The full rank design matrix and parameter vector of a submodel are obtained by dropping those columns from the matrix X above, and those entries from the parameter vector β which correspond to the effects dropped from the full model.

APPENDIX 2

To apply Fisher's method of scoring to maximize the log-likelihood $l(p, \beta)$ one must compute the score

$$U = \left[\frac{\partial l}{\partial p} : \frac{\partial l}{\partial \beta'} \right]' = \left[\frac{\partial l_1}{\partial p} + \frac{\partial l_2}{\partial p} + \frac{\partial l_3}{\partial p} : \frac{\partial l_1}{\partial \beta'} + \frac{\partial l_2}{\partial \beta'} + \frac{\partial l_3}{\partial \beta'} \right]'$$

and the expected information matrix $H = E(UU')$. To simplify the notation, let $x_1 = (x_{1j})$, $y_2 = (y_{2j})$, $z_1 = (z_{1jk})$, $n_1 = (n_{1j})$, $n_2 = (n_{2j})$, $q_1 = (q_{1jk})$, $q_{21} = (q_{2j1})$, $q_{22} = (q_{2j2})$, X_1 the matrix formed by the first 4 rows of X , X_{21} the matrix formed by rows 5 and 7 of X , and X_{22} the matrix formed by rows 6 and 8 of X . Assuming a logit link, that is, $g(q_{ijk}) = \log(q_{ijk}/(1 - q_{ijk}))$, the components of the score vector U are

$$\frac{\partial l_1}{\partial p} = \frac{1}{p(1-p)} 1'(x_1 - n_1 p),$$

$$\frac{\partial l_2}{\partial p} = 0,$$

$$\frac{\partial l_3}{\partial p} = \frac{\partial l_3}{\partial r_2} \cdot \frac{\partial r_2}{\partial p} = (y_2 - n_2 \# r_2)' \text{diag}^{-1}[r_2 \# (1 - r_2)](q_{21} - q_{22}),$$

$$\frac{\partial l_1}{\partial \beta} = 0,$$

$$\frac{\partial l_2}{\partial \beta} = \frac{\partial l_2}{\partial q_1} \cdot \frac{\partial q_1}{\partial \beta} = X_1'(z_1 - m_1 \# q_1), \text{ and}$$

$$\frac{\partial l_3}{\partial \beta} = \frac{\partial l_3}{\partial r_2} \cdot \frac{\partial r_2}{\partial q_2} \cdot \frac{\partial q_2}{\partial \beta} = S' \text{diag}^{-1}[r_2 \# (1 - r_2)](y_2 - n_2 \# r_2).$$

Here $S' = pX'_{21}q_{21}\#(1 - q_{21}) + (1 - p)X'_{22}q_{22}\#(1 - q_{22})$, $\#$ denotes the elementwise product of 2 vectors, and $\text{diag}[v]$ is the diagonal matrix with the elements of the vector v in its diagonal. The expected information matrix is then

$$H = E(UU') = E \begin{bmatrix} \frac{\partial l}{\partial p} \frac{\partial l}{\partial p} & \frac{\partial l}{\partial p} \frac{\partial l}{\partial \beta'} \\ \frac{\partial l}{\partial p} \frac{\partial l}{\partial \beta'} & \frac{\partial l}{\partial \beta} \frac{\partial l}{\partial \beta'} \end{bmatrix}$$

where

$$E \left[\frac{\partial l}{\partial p} \frac{\partial l}{\partial p} \right] = \frac{1}{p(1-p)} 1' n_1 + n_2' \text{diag}^{-1}[\tau_2 \# (1 - \tau_2)](q_{21} - q_{22}),$$

$$E \left[\frac{\partial l}{\partial p} \frac{\partial l}{\partial \beta} \right] = S' n_2 \# (\text{diag}^{-1}[\tau_2 \# (1 - \tau_2)](q_{21} - q_{22})), \text{ and}$$

$$E \left[\frac{\partial l}{\partial \beta} \frac{\partial l}{\partial \beta'} \right] = X_1' \text{diag}[m_1 \# q_1 \# (1 - q_1)] X_1 + S' \text{diag}[n_2] \text{diag}^{-1}[\tau_2 \# (1 - \tau_2)] S.$$

Given starting values p^0 and β^0 for the parameters $\theta = [p, \beta']'$, the maximum likelihood estimates are found iteratively using the updating equations

$$\theta^{n+1} = \theta^n + H^{-1} U.$$

where H and U are evaluated at the current estimate θ^n . Even though the expressions for the score and the information matrix may look complicated, they are easy to program when software for matrix algebra is available (e.g. APL, Gauss, S, SAS-IML). Convergence of the scoring algorithm is rapid, usually in less than five iterations. A SAS (Proc. IML) program to do the computations can be obtained from the author.